# MIND MATRIX

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**SESSION: 05**

**REPORT ON: TASK REPORT**

**TASK NO: 02**

**TITLE:**

**DESIGN A KOTLIN CLASS & DATA MODEL**

## 1. Feature Chosen: Task Tracker App

This feature allows users to manage daily tasks, mark them as completed, and track progress. It improves productivity and organization.

## 2. Data Modeling

**Data Class:**

```
data class Task(
    val id: Int,
    val title: String,
    val description: String,
    var isCompleted: Boolean
)
```

**Interface:**

```
interface TaskManager {
    fun toggleTask(task: Task)
}
```

**Why Data Class?**

- Automatically provides equals(), hashCode(), copy()
- Suitable for immutable data modeling

**Why Interface?**

- Defines behavior abstraction
- Allows flexible implementation

## 3. UML Description

Task (Data Class)

↑

TaskManager (Interface)

↑

TaskViewModel (Implementation Class)

Kotlin features used:

- Primary constructor
- Data class
- Interface
- copy() function

## 4. Mapping to Compose UI

- Tasks can be displayed using LazyColumn.
- When user toggles completion → update state → recomposition occurs.
- Compose observes state changes and refreshes UI automatically.

## 5. Reflection

Modeling behavior required deeper thinking than just creating data fields. Separating data and logic improves maintainability. This structure supports scalability for larger applications.